Complete repair hypergraphs under concatenation

A twisted-cubic–axis family

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Abstract

For a linear code C and a coordinate x, let $\mathcal{H}_x^{(r)}(C)$ be the complete hypergraph of helper sets of size at most r arising from supports of dual words that contain x. Its matching number ν_x is disjoint availability, while its transversal number τ_x is the least number of helper failures that block every radius-r repair. These invariants retain overlap information that locality and availability alone discard.

For $q = 3^h$, we study the $[2q+1, 4, q-1]_q$ code generated by the q affine points of a twisted cubic together with the $q+1$ points of its characteristic-three axis. Axis coordinates have exact locality two and cubic coordinates exact locality three. We determine the axis invariants in terms of the affine cap number, prove $\tau_x = q-2$ with explicit matching bounds at every cubic coordinate, and obtain $\tau_x > \nu_x$ for every coordinate when $q \geq 9$. At $q = 9$ the three coordinate types have exact rows

$$(\nu, \tau) = (4, 7), \qquad (6, 12), \qquad (7, 13).$$

We then prove an exact concatenation theorem. If I is an inner code and $r + 1 < 2d(I^{\perp})$, an outer functional-dual distance of at least $r+2$ forces every concatenated dual word of weight at most $r+1$ into one inner block. Consequently the entire hypergraph $\mathcal{H}_x^{(r)}$, rather than a chosen repair subfamily, is copied block by block. A trace-duality argument turns ordinary extension-field dual distance into this functional gate. Combining the $q = 9$ seed with a self-dual TVZ outer family gives unbounded $\mathbb{F}_9$-linear codes of exact rate $2/19$, eventual relative distance at least $1/5$, exact locality three on the nine cubic coordinates per block, exact locality two on the ten axis coordinates, and the same exact repair rows in every block. Every finite and reduction step has been formalized in Lean. The sole deep import is Stichtenoth's self-dual TVZ theorem, cited with its exact consumed statement.

1 Introduction

Locality asks how many surviving symbols are sufficient to recover one lost coordinate of a code. Availability refines locality by asking for many pairwise-disjoint recovery sets. These parameters have become standard in the theory of locally repairable codes; see, for example, Gopalan–Huang–Simitci–Yekhanin [3] and the geometric constructions of Pamies-Juarez–Hollmann–Oggier [8] and Lavrauw–Van de Voorde [6].

Disjoint availability does not determine robustness against adversarial helper failures. The full family of recovery sets may overlap in a way that admits a small blocking set, or it may remain repairable after every set of failures much larger than a maximum disjoint family. The natural object is therefore the complete bounded repair hypergraph at a coordinate. We write ν for its matching number and τ for its transversal number. Then ν is disjoint availability, while $\tau - 1$ is the maximum number of arbitrary helper failures that can always be tolerated while retaining at least one local repair. Pamies-Juarez–Hollmann–Oggier [8, Definition 3] explicitly introduced this same minimum hitting-set invariant as local repair tolerance, and Wang–Zhang [10] subsequently placed repair tolerance in a general regenerating-set framework. Our emphasis is not the definition of τ, but the exact coordinatewise computation of both ν and τ for a complete code-derived repair family and their simultaneous transfer.

Our inner family comes from a classical object in finite projective geometry. In $\mathrm{PG}(3, q)$, take the affine part of a twisted cubic and, when q has characteristic three, adjoin the common axis of the osculating planes. The axis and its stabilizer orbits are classical; a modern account appears in Davydov–Marcugini–Pambianco [2]. What matters here is not an orbit classification but the complete system of small linear dependencies through each selected point.

The second ingredient is concatenation. Concatenation with a locally repairable inner code is established machinery and yields asymptotically good LRC families; Liu–Ma–Wu–Xing [7], in particular, derive locality and asymptotic bounds this way. Locality preservation only asserts that some inner repairs survive. We isolate a stronger dual-distance condition under which *every* bounded repair support in the concatenated code is an embedded inner repair support. Thus matching, transversal, edge counts, and all other invariants of the complete repair hypergraph transfer exactly.

Contributions

The paper proves the following package.

(i) We construct a characteristic-three code C_q with parameters $[2q+1, 4, q-1]_q$ and classify every circuit of size at most four.

(ii) We compute exact axis-coordinate matching and transversal numbers in terms of the affine cap number, prove exact cubic-coordinate transversals and matching bounds, and deduce $\tau_x > \nu_x$ for every coordinate when $q \geq 9$.

(iii) We give the complete $q = 9$ table, including the three exact (ν, τ) rows and the counts of inclusion-minimal repairs.

(iv) We prove exact preservation of complete radius-r repair hypergraphs under the block-confinement hypothesis $r + 1 < 2d(I^{\perp})$ and an outer functional-dual distance gate.

(v) We prove a finite-separable trace bridge from ordinary extension-field dual distance to that functional gate.

(vi) Using Stichtenoth's self-dual TVZ family [9], we derive a fixed-alphabet family over $\mathbb{F}_9$ with exact rate $2/19$, eventual relative distance at least $1/5$, exact mixed locality, and exact blockwise repair rows.

Novelty posture

The twisted cubic, its characteristic-three axis, linear-code locality, multiple repair alternatives, and locality-preserving concatenation are prior art. So are repair tolerance as a minimum hitting set [8], general regenerating-set formulations [10], and refined dual-support distributions that capture locality [4]. The candidate contributions are narrower: the exact all-symbol matching/transversal separation for this point system and equality of the *complete* bounded repair hypergraph under the stated outer-dual gate. A targeted search found adjacent work on twisted-cubic incidence and covering counts [1], geometric availability [8, 6], and concatenated locality [7, 5], but did not locate either statement. This is a bounded search result, not a priority certificate; the accompanying novelty ledger records the exact search boundary and the paper avoids unconditional "first" language.

2 Complete bounded repair hypergraphs

Let $C \le \mathbb{F}_q^X$ be a linear code on a finite coordinate set X, with dual $C^\perp$ under the standard dot product.

Definition 2.1. Fix $x \in X$ and $r \ge 0$. The *complete radius-r repair hypergraph* at x is

$$\mathcal{H}_x^{(r)}(C) = \left\{ R \subseteq X \setminus \{x\} : |R| \le r, \begin{array}{l} \text{there is } y \in C^\perp \text{ with } y_x \ne 0, \\ \operatorname{supp}(y) = \{x\} \cup R \end{array} \right\}.$$

Its inclusion-minimal members form the repair clutter $\mathcal{H}_{x,\min}^{(r)}(C)$. We put

$$\nu_x^{(r)}(C) = \nu(\mathcal{H}_x^{(r)}(C)), \qquad \tau_x^{(r)}(C) = \tau(\mathcal{H}_x^{(r)}(C)),$$

where ν is maximum matching size and τ is minimum transversal size.

The word y supplies the recovery equation

$$c_x = -y_x^{-1} \sum_{z \in R} y_z c_z \qquad (c \in C).$$

Thus Definition 2.1 includes every linear repair whose coefficient support has at most r helpers; it is not a declared or selected subfamily.

Proposition 2.2. *For every finite code-derived repair hypergraph with nonempty edges,*

$$\nu_x^{(r)}(C) \le \tau_x^{(r)}(C).$$

Passing to the inclusion-minimal repair clutter preserves both ν and τ. If $r + 1 \le d(C^\perp)$, every edge has exactly r helpers.

Proof. A transversal meets every edge of a matching, and the matching edges are disjoint, proving $\nu \le \tau$. Every edge contains an inclusion-minimal edge. Consequently a set hits all edges exactly when it hits all minimal edges; replacing each matching edge by a minimal subedge preserves disjointness. Finally, an edge R is witnessed by a nonzero dual word of weight $|R| + 1$. Dual distance gives $r + 1 \le |R| + 1$, while the definition gives $|R| \le r$. $\square$

The operational meaning of τ is immediate: every failure set of size at most $\tau - 1$ misses some repair edge, whereas a transversal of size τ blocks them all. Matching and transversal can nevertheless be far apart; equality is not a generic hypergraph law.

3 The characteristic-three twisted-cubic–axis code

Let $q = 3^h$ and write $\mathbb{F} = \mathbb{F}_q$. In homogeneous coordinates on $\mathrm{PG}(3, q)$ define

$$T_q = \{C(t) = (1 : t : t^2 : t^3) : t \in \mathbb{F}\}$$

and

$$L_q = \{A(u) = (0 : 1 : u : 0) : u \in \mathbb{F}\} \cup \{A_\infty = (0 : 0 : 1 : 0)\}.$$

The set T_q is the affine twisted cubic and L_q is the full line $X_0 = X_3 = 0$. For $q \equiv 0 \pmod 3$, this is the common axis of the osculating-plane pencil [2, Section 7]. We use that geometric name only as provenance: all proofs below start from the displayed coordinates.

Let $S_q = T_q \sqcup L_q$ and let G_q be the $4 \times (2q + 1)$ matrix whose columns are fixed representatives of these points. Write C_q for the row code of G_q.

Theorem 3.1 (Code parameters)**.** *For every finite field of characteristic three,*

$$C_q = [2q+1, 4, q-1]_q.$$

Proof. The length is $2q+1$, and four displayed columns span $\mathbb{F}^4$. The distance of a projective-system code is its length minus the largest hyperplane section. If a plane does not contain L_q, it meets L_q in at most one point and its equation restricts to a nonzero cubic polynomial on T_q, so it contains at most three points of T_q. If it contains L_q, its equation has the form $aX_0 + dX_3 = 0$. On T_q this becomes $a + dt^3 = 0$, which has at most one solution because Frobenius $t \mapsto t^3$ is injective. Hence every plane section has size at most $q+2$. The plane $X_3 = 0$ contains all of L_q and the point $C(0)$, attaining $q+2$. Therefore $d = (2q+1) - (q+2) = q-1$. $\square$

3.1 Small circuits and exact locality

For distinct $s, t, u \in \mathbb{F}$, put

$$e_1 = s + t + u, \qquad e_2 = st + su + tu.$$

Theorem 3.2 (Small-circuit classification)**.** *The inclusion-minimal dependent subsets of S_q having size at most four are exactly:*

(a) the triples of distinct points of L_q;

(b) the sets

$$\{C(s), C(t), C(u), (0 : e_1 : e_2 : 0)\}$$

for distinct $s, t, u \in \mathbb{F}$.

The pair (e_1, e_2) is never $(0, 0)$ for distinct s, t, u, so the axis completion in (b) is a unique projective point. Consequently every axis coordinate has exact locality two and every cubic coordinate has exact locality three.

Proof. Every three distinct points on L_q are minimally dependent. Vandermonde independence handles up to four cubic columns before the point at infinity is punctured, and direct 4×4 determinants exclude the mixed types with two cubic and two axis columns. Expanding the determinant of three cubic columns and $(0 : e_1 : e_2 : 0)$ gives zero. Conversely, the same determinant shows that any dependent three-cubic/one-axis set has this axis point. If $e_1 = e_2 = 0$, then s, t, u would be the three roots of $X^3 - stu$; injectivity of Frobenius would force them equal. Thus the four points form a circuit. The locality assertions follow from the dual-support/circuit correspondence and the absence of smaller circuits through the respective target type. $\square$

4 Matching and transversal invariants

Let $Z_3(q)$ denote the largest size of a subset of the additive group of $\mathbb{F}_q$ containing no three distinct elements with sum zero. Under an $\mathbb{F}_3$-basis, this is the usual affine cap number in $\mathrm{AG}(h, 3)$.

4.1 Axis coordinates

Fix $y \in L_q$. The minimal radius-three repair clutter at y splits over two disjoint helper grounds:

- every pair among the q other axis points repairs y, giving the complete graph K_q;
- a cubic triple repairs y exactly when its completion in Theorem 3.2 equals y.

Matching and transversal numbers add across this disjoint union.

For A_∞, the cubic triples are exactly the distinct zero-sum triples of $\mathbb{F}_q$, hence the affine lines of $\mathrm{AG}(h,3)$. For $A(a)$, the permutation

$$t \longmapsto (t+a)^{-1}, \qquad 0^{-1} = 0,$$

identifies the cubic repair component with the zero-sum triples avoiding zero.

Theorem 4.1 (Exact axis rows)**.** *At the nucleus-type point A_∞,*

$$\nu_{A_\infty} = \frac{5q-3}{6}, \qquad \tau_{A_\infty} = 2q - 1 - Z_3(q).$$

At every finite axis point $A(a)$,

$$\nu_{A(a)} = \frac{5q-9}{6}, \qquad \tau_{A(a)} = 2q - 2 - Z_3(q).$$

These are the invariants of both the minimal repair clutter and the complete radius-three repair hypergraph.

Proof. The complete graph K_q contributes matching $(q-1)/2$ and transversal $q-1$. The affine-line hypergraph contributes matching $q/3$ and transversal $q - Z_3(q)$: parallel lines give the matching, and complements exchange transversals with zero-sum-free sets. After forbidding zero, the corresponding values are $q/3 - 1$ and $q - 1 - Z_3(q)$. Adding the contributions gives the formulas. Proposition 2.2 passes the invariants between the complete hypergraph and its minimal clutter. □

The strict gap does not require a quantitative cap-set bound. Any matching in the minimal axis clutter consumes two axis helpers for each pair edge and three cubic helpers for each cubic edge. Weighting axis helpers by three and cubic helpers by two yields $6\nu_y \leq 5q$. The K_q component alone forces $\tau_y \geq q-1$. Thus $\nu_y < \tau_y$ for $q \geq 9$.

4.2 Cubic coordinates

Fix $x \in \mathbb{F}$. By Theorem 3.2, every unordered pair $\{s,t\} \subseteq \mathbb{F} \setminus \{x\}$ gives one repair

$$\{C(s), C(t), A(\kappa_x(s,t))\},$$

where $\kappa_x(s,t)$ is the unique completion color. For fixed s, the map $t \mapsto \kappa_x(s,t)$ is injective away from the diagonal. The repair hypergraph is therefore a properly edge-colored K_{q-1} in which every graph edge is augmented by its color vertex on the axis.

Theorem 4.2 (Cubic rows)**.** *Every cubic coordinate has exactly $\binom{q-1}{2}$ radius-three repairs and*

$$\frac{q}{3} \leq \nu_x \leq \frac{q-1}{2}, \qquad \tau_x = q - 2.$$

In particular, $\tau_x > \nu_x$ for $q \geq 9$.

Proof. The edge count is the number of unordered pairs of cubic helpers. A matching uses two distinct cubic helpers per edge, giving the upper bound. A maximal rainbow-matching argument in the properly colored K_{q-1} gives $\lceil (q-2)/3 \rceil = q/3$ disjoint augmented edges.

For the transversal upper bound, choose one cubic helper $a \neq x$ and take all other cubic helpers except x and a; this set has size $q-2$ and every repair uses one of them. Conversely, suppose a transversal leaves a nonempty set U of cubic helpers unchosen. Fix $a \in U$. For each $b \in U \setminus \{a\}$, the edge on $\{a, b\}$ forces its completion color into the transversal, and properness makes these colors distinct. Counting the chosen cubic helpers together with these forced colors gives at least $q-2$ vertices. □

Combining the two coordinate types gives the uniform statement.

Corollary 4.3 (All-symbol separation)**.** *For every $q = 3^h \geq 9$ and every coordinate z of C_q,*

$$\nu_z^{(3)}(C_q) < \tau_z^{(3)}(C_q).$$

Moreover, C_q has all-symbol locality three, with exact locality two on L_q and exact locality three on T_q.

4.3 The exact field-nine seed

For $q = 9$, the affine cap number is $Z_3(9) = 4$. The cubic matching bound is also sharp. We obtain the complete table below; repair counts refer to inclusion-minimal repairs, while (ν, τ) is unchanged for the complete hypergraph.

coordinate type	multiplicity	locality	minimal repairs	ν	τ
cubic $C(x)$	9	3	28	4	7
finite axis $A(a)$	9	2	$36 + 8$	6	12
axis A_∞	1	2	$36 + 12$	7	13

Table 1: Exact rows of the $[19, 4, 8]_9$ seed.

The 36 pair repairs are the edges of K_9. The cubic components contain eight zero-sum triples avoiding zero at a finite axis point and twelve zero-sum triples at A_∞. The small-circuit inventory consists of $\binom{10}{3} = 120$ axis triples and $\binom{9}{3} = 84$ completed cubic four-circuits.

Corollary 4.4. *Every coordinate of $C_9 = [19, 4, 8]_9$ satisfies*

$$7\nu_z \leq 4\tau_z.$$

The constant $7/4$ is attained at every cubic coordinate.

5 Exact repair transfer under concatenation

We formulate the transfer theorem in a coordinate-free way. Let $I \leq \mathbb{F}^K$ be an inner code and let V be an $\mathbb{F}$-vector space equipped with a linear isomorphism $e : V \to I$. For an outer $\mathbb{F}$-linear code $O \leq V^J$, the concatenated code $O \circ_e I \leq \mathbb{F}^{J \times K}$ encodes each outer symbol through e.

The *functional dual* of O is

$$O^{\perp_{\text{fun}}} = \left\{ (\beta_j)_{j \in J} \in (V^*)^J : \sum_{j \in J} \beta_j(u_j) = 0 \text{ for every } u \in O \right\}.$$

Its distance is the minimum number of nonzero component functionals in a nonzero tuple. This is intrinsic to the symbol space V and does not choose field coordinates.

Theorem 5.1 (Complete repair-hypergraph transfer)**.** *Let $r \geq 0$. Suppose*

$$r+1 < 2d(I^{\perp}) \quad \textit{and} \quad d(O^{\perp_{\mathrm{fun}}}) \geq r+2.$$

Then for every block $j \in J$ and inner coordinate $x \in K$,

$$\mathcal{H}^{(r)}_{(j,x)}(O \circ_e I) = \{\{j\} \times R : R \in \mathcal{H}^{(r)}_x(I)\}.$$

Consequently matching number, transversal number, inclusion-minimal repair clutter, edge counts, and the full incidence structure are preserved coordinate by coordinate.

Proof. Take $w \in (O \circ_e I)^{\perp}$ with $\mathrm{wt}(w) \leq r+1$ and write w_j for its blocks. Each w_j defines a functional $\beta_j \in V^*$ by pairing w_j with $e(v)$. Orthogonality to the concatenated code says $(\beta_j)_j \in O^{\perp_{\mathrm{fun}}}$. Its functional support has size at most $\mathrm{wt}(w) \leq r+1$, below the outer functional-dual distance. Hence every β_j is zero, so each block w_j lies in $I^{\perp}$.

Every nonzero block now has weight at least $d(I^{\perp})$. If two blocks were nonzero, then

$$\mathrm{wt}(w) \geq 2d(I^{\perp}) > r+1,$$

a contradiction. Thus a bounded dual word containing (j,x) is supported only in block j, where its helper support is an inner repair of x. This proves one inclusion. Conversely, extending any inner dual word by zero outside block j gives a dual word of every concatenation, proving the other inclusion. ☐

The inequality $r+1 < 2d(I^{\perp})$ is important. It does not require $d(I^{\perp}) = r+1$. For the full $q=9$ seed, $d(C_9^{\perp}) = 3$ because of the axis triples, yet radius-three transfer is exact since $4 < 2 \cdot 3$.

6 The trace bridge and the field-nine lift

Let $L/\mathbb{F}$ be a finite separable extension and let $O \leq L^J$ be an L-linear outer code. Restricting scalars makes each outer symbol an $\mathbb{F}$-vector space. The functional-dual gate in Theorem 5.1 is then supplied by the trace pairing.

Theorem 6.1 (Trace-duality bridge)**.** *Every $\mathbb{F}$-linear functional $f : L \to \mathbb{F}$ has a unique representation*

$$f(x) = \mathrm{Tr}_{L/\mathbb{F}}(a_f x).$$

If $(f_j)_j$ belongs to the functional dual of the restricted-scalar code, then $(a_{f_j})_j \in O^{\perp}$. Moreover

$$f_j = 0 \quad \Longleftrightarrow \quad a_{f_j} = 0$$

at every coordinate. Hence

$$d((O|_{\mathbb{F}})^{\perp_{\mathrm{fun}}}) \geq d(O^{\perp}).$$

Proof. Separable trace is a nondegenerate bilinear pairing, giving the unique coefficient a_f and exact support preservation. If $u \in O$ and $c \in L$, functional orthogonality applied to $cu \in O$ gives

$$0 = \sum_j f_j(cu_j) = \mathrm{Tr}_{L/\mathbb{F}}\left(c \sum_j a_{f_j} u_j\right) \qquad \text{for every } c \in L.$$

Nondegeneracy forces $\sum_j a_{f_j} u_j = 0$. Thus the coefficient tuple lies in the ordinary L-dual of O, with the same support. ☐

Now take $\mathbb{F} = \mathbb{F}_9$, $L = \mathbb{F}_{9^4} = \mathbb{F}_{6561}$, and use a linear equivalence between the four-dimensional $\mathbb{F}_9$-space L and the inner code C_9.

Corollary 6.2 (Finite extension-field lift)**.** *Let $O \le L^N$ be an $[N, K, D]_L$ code with $O \ne 0$ and $d(O^\perp) \ge 5$. Its concatenation with C_9 is an $\mathbb{F}_9$-linear code with*

$$[19N,\ 4K,\ \ge 8D]_9.$$

Its complete radius-three repair hypergraph is an embedded copy of the corresponding C_9 hypergraph in each block. Consequently, among its $19N$ coordinates, $9N$ cubic coordinates have exact locality three and row $(\nu, \tau) = (4, 7)$, $9N$ finite-axis coordinates have exact locality two and row $(6, 12)$, and N infinity-axis coordinates have exact locality two and row $(7, 13)$. Their guaranteed arbitrary-helper failure thresholds $\tau - 1$ are respectively 6, 11, and 12.

Proof. Restriction of scalars multiplies dimension by four and does not change symbol support. The usual concatenated-distance argument gives distance at least $8D$. Theorem 6.1 supplies functional-dual distance at least five, while $4 < 2d(C_9^\perp) = 6$. Apply Theorem 5.1 and Table 1. □

7 A fixed-alphabet asymptotic family

We use one deep existence theorem and expose it completely. Stichtenoth [9, Theorem 1.6(ii)] proves that for every square $Q = \ell^2$ there is an unbounded sequence of self-dual $[N_j, N_j/2, D_j]_Q$ codes with

$$\liminf_j \frac{D_j}{N_j} \ge \frac{1}{2} - \frac{1}{\ell - 1}.$$

For $Q = 6561 = 81^2$, the lower bound is $39/80$. Self-duality is especially useful here: the same D_j supplies both the primal distance needed for the concatenated distance and the dual distance needed for exact repair transfer.

Theorem 7.1 (Uniform asymptotic repair family)**.** *There is a sequence of $\mathbb{F}_9$-linear codes $\mathcal{C}_j$ with lengths tending to infinity such that, for all sufficiently large j,*

(a) $\mathcal{C}_j$ has exact rate $2/19$;

(b) its relative minimum distance is at least $1/5$;

(c) in every inner block, nine coordinates have exact locality three and ten have exact locality two;

(d) according to its inner coordinate type, its complete radius-three repair hypergraph has exact row $(4, 7)$, $(6, 12)$, or $(7, 13)$.

In particular $7\nu_x \le 4\tau_x$ at every coordinate.

Proof. Take Stichtenoth's self-dual codes over $L = \mathbb{F}_{6561}$ and concatenate them with C_9 after restriction to $\mathbb{F}_9$. Since $39/80 > 1/3$, eventually

$$N_j \le 3D_j \quad \text{and} \quad N_j \ge 15.$$

The second inequality and the first imply $D_j \ge 5$, so self-duality gives $d(O_j^\perp) = D_j \ge 5$ and Corollary 6.2 applies. Since also $39/80 > 19/40$, eventually $19N_j \le 40D_j$. The lifted length and dimension are $19N_j$ and $4(N_j/2) = 2N_j$, giving exact rate $2/19$. Its distance is at least $8D_j$, and

$$\frac{d(\mathcal{C}_j)}{19N_j} \ge \frac{8D_j}{19N_j} \ge \frac{1}{5}.$$

The exact localities and repair rows transfer blockwise. □

Remark 7.2 (Meaning of "unconditional"). Theorem 7.1 is unconditional as ordinary mathematics once Stichtenoth's published theorem is accepted. In the Lean trust ledger it is not a zero-project-axiom theorem: the exact specialization of Theorem 1.6(ii) is the single quarantined literature axiom. Every reduction from that statement to Theorem 7.1 is kernel-checked.

8 Relation to prior work

The locality of codeword symbols and its parameter bounds originate in the modern LRC literature represented by [3]. Multiple repair alternatives and finite-geometric constructions are explicit in Pamies-Juarez–Hollmann–Oggier [8]; exact availability for large classes of generalized quadrangles is developed in Lavrauw–Van de Voorde [6]. In particular, [8, Definition 3] defines local repair tolerance by the same minimum hitting-set formula as our τ, while Wang–Zhang [10] organize several tolerance notions through regenerating sets. Thus neither repair tolerance nor the hypergraph interpretation is claimed as new here. The candidate contribution is the exact coordinatewise pair (ν, τ), its strict separation on every symbol of this family, and the transfer of the entire bounded repair family.

Gruica–Jany–Ravagnani [4] give a refined weight distribution indexed by prescribed subsets of dual-word supports and prove duality identities for it. This is close prior art for treating all bounded dual supports rather than a selected repair group. It does not state the matching/transversal computations below or a concatenation theorem that identifies the complete bounded support family.

Concatenation is likewise not new. Liu–Ma–Wu–Xing [7, Theorem 3.1] concatenate a locally repairable inner code with classical outer codes and derive locality and asymptotic parameter families. The recent construction of Jin–Fu [5] also uses concatenation to obtain binary LRCs with disjoint local repair groups and controlled weight distributions. These results establish locality and code parameters, not equality of every bounded repair support. Our transfer theorem strengthens the conclusion under an additional outer-dual gate: it identifies every bounded dual support and therefore transfers the complete repair hypergraph, not only the existence of an inner recovery set.

The geometry of the twisted cubic and its stabilizer has a substantial literature. Davydov–Marcugini–Pambianco [2] describe the axis in characteristic three and line orbits under the cubic stabilizer; Bartoli–Davydov–Marcugini–Pambianco [1] compute point–plane incidence data and multiple-covering-code parameters. Those incidence counts are adjacent to, but do not by themselves determine, matching and transversal numbers of the repair hypergraphs in Section 4.

The exact novelty boundary is maintained outside the manuscript in a claim-by-claim ledger and an adversarial search report. The intended claim is not that twisted-cubic codes, availability, trace duality, or concatenation are new. It is the conjunction of an explicit all-symbol $\tau > \nu$ seed with exact complete-hypergraph transfer and concrete fixed-alphabet parameters. Until a specialist citation-chain review is complete, this should be read as a candidate contribution supported by a documented none-found search, not as a categorical priority statement.

9 Formal verification and provenance

The project was human-directed and agent-generated. The named author selected the research program and publication policy; AI systems generated substantial portions of the mathematics, code, formal proofs, and prose. AI is not listed as an author. This section makes the verification route visible rather than relegating it to acknowledgments.

The complete finite theorem chain is formalized in Lean 4 using mathlib. The formal objects are ordinary finite-dimensional linear codes, dual-word supports, finite hypergraphs, matching and transversal numbers, finite fields, restriction of scalars, and the algebra trace. The formal proofs establish:

- the code parameters and small-circuit classification;
- exact locality and all matching/transversal statements;
- the complete repair-hypergraph equality under concatenation;
- the finite-separable trace bridge with exact support;
- the concrete fields $\mathbb{F}_9 \subset \mathbb{F}_{6561}$ and extension degree four;
- the disjoint exhaustive lifted coordinate-type partition and its exact multiplicities $9N, 9N, N$;
- the analytic extraction of $N \leq 3D$ and $19N \leq 40D$ from the imported limit bound;
- the final constants 2/19 and 1/5.

No `sorry`, `admit`, `native_decide`, `unsafe`, or code-generation trust occurs in this proof chain. The finite checks use kernel reduction. Axiom reports for all finite, algebraic, and analytic-reduction declarations contain only `propext`, `Classical.choice`, and `Quot.sound`. The asymptotic headline adds exactly one project axiom, `Imported.stichtenoth_selfDual_TVZ_6561`, whose docstring cites Stichtenoth's theorem number, specialization, and exact parameter statement.

The machine-readable trust manifest is `lean/RepairCodes/TRUST.md`. The paper package includes a separate proof ledger mapping every displayed theorem to Lean declarations and an adversarial novelty review separating mathematical novelty from formalization novelty.

A Statement-adequacy excerpts

This appendix prints the load-bearing Lean declarations in their source form. Typeclass hypotheses—finite fields, finite index types, decidable equality, and finite-dimensionality—are declared immediately above these statements in the source files.

Exact complete-repair transfer

```
theorem repairHypergraph_concatenatedCode_eq_embed
    (I : Submodule F (kappa -> F)) (e : V ~=l[F] I)
    (O : Submodule F (iota -> V)) (r : Nat)
    (hsI : r + 1 < 2 * dualDist I)
    (hdO : HasFunctionalDualDistanceAtLeast O (r + 2))
    (j : iota) (x : kappa) :
    repairHypergraph (concatenatedCode I e O) (j, x) r =
      embedHypergraph (blockEmbedding j) (repairHypergraph I x r)
```

The ASCII substitutions `F`, `kappa`, `iota`, and `≅l` above are typographical transliterations of the Unicode identifiers in the checked source; the proposition is otherwise verbatim.

Trace-duality bridge

```
theorem hasFunctionalDualDistanceAtLeast_restrictScalars
    (O : Submodule L (iota -> L)) (d : Nat)
    (hd : d <= dualDist O) :
    HasFunctionalDualDistanceAtLeast (O.restrictScalars K) d
```

Exact extension lift

```
theorem q9ExtensionLiftCode_parameters
    (hcard : Fintype.card F = 9)
    (hdeg : Module.finrank F L = 4)
    (O : Submodule L (iota -> L)) (hO : O != bottom) :
    Fintype.card (iota x AxisTwistedCubicIndex F) =
        19 * Fintype.card iota /\
    Module.finrank F (q9ExtensionLiftCode hdeg O) =
        4 * Module.finrank L O /\
    8 * minDist O <= minDist (q9ExtensionLiftCode hdeg O)
```

Concrete asymptotic headline

```
theorem concrete_q9_uniform_repair_family :
    HasQ9UniformRepairFamily finrank_q9OuterField :=
  stichtenoth_q9_uniform_repair_family
    card_q9BaseField card_q9OuterField finrank_q9OuterField
```

Here `HasQ9UniformRepairFamily` expands to unbounded length, eventual $19k = 2n$, eventual $n \leq 5d$, exact locality three/two according to inner coordinate type, and the exact three-way matching/transversal row match displayed in Table 1. The complete definition is recorded in `RepairCodes/Asymptotic.lean`; the external proof ledger points to the exact source line and axiom report.

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